TASK 1 — Understand the Problem and Identify the Dat

1. Problem Analysis

The Grama Jyothi Rural Electricity Co-operative supplies electricity to households. The billing program calculates electricity consumption from the previous and current meter readings and determines the total bill using the C PROGRAMMING — ASSIGNMENT 1

Assignment Title: Design, Implement and Validate a Reliable Billing Program for a Rural Electricity Co-operative

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Course: C Programming

Submission Date: 26/applicable tariff rates and fixed charge.

The program must handle meter rollover, avoid negative bills, calculate monetary values accurately to the nearest paisa, and support a large number of households.

2. Inputs and Outputs

Inputs:

- Previous meter reading
- Current meter reading
- Household number
- Consumer category
- Fixed charge and applicable tariff rates

Outputs:

- Electricity units consumed

- Energy charge

- Fixed charge

- Total bill payable

2.1 Justification of Each Data Item

1. Previous and current meter readings: These are whole numbers from 0 to 65535. An unsigned integer type such as "uint32_t" is suitable because negative readings are invalid.

2. Household number: An integer type is suitable for identifying each household. If the identifier contains letters or leading zeros, a character string should be used instead.

3. Consumer category: An "enum" is suitable because the consumer category is selected from a limited set of predefined options.

4. Units consumed: An unsigned integer type such as "uint64_t" can store large consumption totals without negative values.

5. Tariff rates and fixed charge: Store monetary values in integer paise to avoid floating-point rounding errors. For example, ₹2.25 is stored as 225 paise.

6. Energy charge and total bill: Use a sufficiently wide integer type to store the amount in paise. Validate calculations to prevent overflow.

7. Constants: The meter modulus (65536), free-unit limit (100), and tariff rates should be defined as named constants so they can be updated consistently.

8. Inputs that may be invalid: Meter readings must be checked against the range 0–65535. Invalid readings must be rejected before calculating the bill.

3. Constants and Variables

Constants include the meter maximum reading (65535), meter modulus (65536), the first 100 free units, and the specified tariff rates.

Variables include meter readings, household number, consumer category, units consumed, energy charge, fixed charge, and total bill.

Money should be stored as integer paise where practical to avoid floating-point rounding errors.

4. Billing Rules

1. Meter readings must be between 0 and 65535.

2. If the current reading is greater than or equal to the previous reading, consumption is their difference.

3. If the meter rolls over, consumption must be calculated using the meter modulus

Example: Meter Rollover

- Previous reading : 65530
- Current reading :5
- Meter Modulus:65536

CALCULATION:

(65536-65530)+5=11

Therefore ,electricity consumption is 11 units

4. The first 100 units are free under the given example tariff.

5. Units 101–200 cost ₹2.25 per unit.

6. Units above 200 cost ₹4.50 per unit.

7. The total bill equals the energy charge plus the fixed charge.

8. Invalid inputs and arithmetic overflow must be handled safely.

5. Assumptions

- The meter is a 16-bit meter with readings from 0 to 65535.
- The given tariff applies unless the assignment specifies a different consumer category.
- The fixed charge must be supplied because its amount is not stated in the scenario.
- The first 100 units are free under the example tariff.

6. Alternative Concept and Complexity Analysis

If meter readings are stored as text, such as "01530", instead of numeric values, the program must validate the characters and convert the text into numbers before performing arithmetic.

For numeric readings, subtraction takes O(1) time and O(1) space for fixed-width integers. For text readings containing d digits, conversion takes O(d) time and O(d) space when the input string is stored.

7. Conclusion

Using suitable integer data types, integer paise for monetary calculations, named constants, and input validation will improve the accuracy, reliability, and maintainability of the billing program.